

Democracy Dies in Darkness

Your baby's sex isn't random. A study shows what could influence it.

A new study finds that the odds of having a boy or girl aren't 50-50. Instead, birth sex is associated with maternal age, specific genes and the sexes of older siblings.

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By [Sabrina Malhi](#)

A baby's sex may not be up to mere chance.

A study published Friday in the journal *Science Advances* describes the odds of having a boy or girl as akin to flipping a weighted coin, unique to each family. It found evidence that an infant's birth sex is associated with maternal age and specific genes.

The findings challenge assumptions that birth sex is random. They mirror the results of similar studies in Europe that have also found that birth sex does not follow a simple 50-50 distribution.

Scientists have long documented a global imbalance in which slightly more boys are born than girls. The new study examined the murkier patterns of birth sex within individual families.

To do so, researchers at the Harvard T.H. Chan School of Public Health analyzed data from more than 146,000 pregnancies from 58,000 nurses in the United States between 1956 and 2015. They found that some families were more likely to have children of the same sex than would be expected if each baby had an equal chance of being a boy or a girl. Moms with three or more kids were more likely to have all boys or all girls than expected by chance.

The study suggests that sex at birth follows a weighted probability and that biological influences may sway the sex of the child.

"If you've had two girls or three girls and you're trying for a boy, you should know your odds are not 50-50," said Jorge Chavarro, the study's senior author. "You're more likely than not to have another girl."

Researchers estimated that families with three girls had a 58 percent chance of having another girl, while families with three boys had a 61 percent chance of having a fourth boy.

Maternal age is a key factor. Women who started having children after age 28 were slightly more likely to have only boys or only girls. Chavarro said this could reflect biological changes as women age, such as increased vaginal acidity, that influence the survival of the Y chromosome carried by boys. Paternal factors could also play a role because maternal and paternal ages are often closely linked. But the study did not include data on fathers, which was noted as a limitation.

Researchers also identified two genes associated with giving birth to only boys or only girls. “We don’t know why these genes would be associated with sex at birth, but they are, and that opens up new questions,” said Chavarro, a professor of nutrition and epidemiology.

Iain Mathieson, a professor of genetics at the University of Pennsylvania Perelman School of Medicine who was not involved in the study, said the genetic associations should be viewed cautiously. He said the study’s genetic analysis was based on a relatively small sample and may be influenced by other factors, making the results more speculative until confirmed by further research.

“I don’t find the genetic factors identified here particularly convincing,” Mathieson said in an email.

The researchers also found that parents were more likely to have one boy and one girl than would be expected by chance, a pattern they believe reflects a tendency to stop having children once both sexes are represented.

To reduce bias from such family planning decisions, they analyzed data after removing each woman’s final child. They also excluded women who had experienced miscarriages or stillbirths to test whether pregnancy loss changed the results. They still found the same pattern: The odds of birth sex did not follow mere chance.

The study suggests that it may not have been so improbable for the fictional parents in the TV sitcom “Malcolm in the Middle” to have five sons or for the Bennet family in “Pride and Prejudice” to have five daughters. Even in history, patterns like this have drawn attention. King Louis VII of France, for example, who reigned in the 12th century, remarried after his first two wives each gave birth to daughters, depriving him of a male heir.

Chavarro said it might take years to fully understand why some families consistently have children of one sex, but this research is an important place to start.

His team said future studies should explore how lifestyle, nutrition and exposure to environmental chemicals might affect these patterns.

Certain factors such as race, natural hair color, blood type, body mass index and height were not associated with having children of only one sex. But the study sample was 95 percent White and made up entirely of nurses, a group that may have different occupational exposures or health patterns compared with the general population.

David A. Haig, an evolutionary biologist at Harvard University who was not involved in the study, said it offers evidence that the probability of a baby's sex varies by family.

"Different families are flipping different coins with different biases," Haig said. "It speaks to something very intuitive and personal, even if the underlying biology is complex."
